

Hackathon Challenge: AI Evaluation Engine for Elevate

Problem Statement

How can Elevate objectively measure the quality of AI-generated learning?

Elevate uses AI to create personalised learning content and practice experiences for users. There are guardrails and frameworks embedded in the current system to ensure outputs generated do not deviate.

Since AI is non deterministic and hyper personalization inherently means many different scenarios can be input to the content generation module, an important engineering challenge emerges:

How do we know whether an AI-generated response is actually good?

An AI response can be fluent, well-written and convincing while still being incomplete, inaccurate, irrelevant, too difficult for the learner, or poorly suited to the intended skill.

Today, evaluating such responses can involve subjective human judgement. Elevate needs a systematic and scalable way to evaluate AI outputs using clearly defined criteria and evidence.

Your challenge is to build an **AI Evaluation Engine** that can objectively assess the quality of AI-generated learning and/or practice responses.

The Challenge

Build a reusable evaluation system that takes an Elevate class scenario and the learning content generated to analyze:

- How relevant is the response to the user?
- What dimensions does it perform well or poorly on?
- Why did it receive that score?
- What evidence from the response supports the evaluation?
- How consistent is the evaluation?

The system should work beyond a single hard-coded example.

Teams should design a framework that can be adapted to different Elevate learning modules.

Example

Consider an Elevate learning scenario [This can be accessed on <https://elevate-skill.com> → Class → My Lessons → Click on Decision Making]:

User scenario:

"I have to give a presentation to my college class. I get nervous and speak too quickly. Teach me how to improve my delivery."

An AI-generated response might explain techniques such as:

- slowing down
- using pauses
- varying vocal tone
- practising beforehand
- recording yourself

The question is:

Is this a good response?

A simple LLM score of "8/10" isn't enough.

The evaluation system should be able to determine, for example [These are indicative only, the team should come up with a comprehensive list of parameters for evaluation]:

Dimension	Score	Reason
Relevance	9/10	Directly addresses presentation delivery
Accuracy	9/10	Recommendations are consistent with established communication practices
Completeness	7/10	Does not sufficiently address nervousness
Clarity	8/10	Easy for the intended learner to understand
Actionability	7/10	Provides techniques but limited practice instructions

Personalisation 6/10 Uses the scenario but provides limited tailoring

The system should also provide **evidence and reasoning** for these scores.

What Should the Evaluation Engine Measure?

The exact evaluation dimensions are part of the challenge and the teams should come up with their own dimensions.

Indicative dimensions as a suggestion for teams include:

Relevance

Does the response actually address the user's question, scenario or task?

Accuracy

Is the information factually correct and free from misleading claims?

Completeness

Does the response cover the important elements expected for the task?

Clarity

Can the intended learner easily understand the response?

Actionability

Does the response give the learner something useful that they can apply or practise?

Personalisation

Does the response appropriately use the context provided by the user?

Structure

Is the information organised logically and presented in a useful sequence?

Level Appropriateness

Is the complexity appropriate for the learner's age, experience and stated context?

Human Likeness Score:

Does the response sound like something a person would naturally say?

Coherence:

Does the response flow logically or there breaks from one scene to the next.

Expected Solution

The expected solution is a working **AI Evaluation Engine**, rather than simply a prompt that asks an LLM to provide a score.

At a high level:

Scenario / Task

↓

AI-generated response

↓

Evaluation Engine

↓

Evaluation Rubric

↓

Dimension Scores

↓

Overall Score + Evidence + Explanation

The exact architecture and implementation are open to the teams.

Key Requirements

Requirement 1 — Multi-dimensional Evaluation

The system should evaluate responses across multiple meaningful dimensions rather than producing only one overall score.

Teams should explain why each dimension was selected.

Requirement 2 — Evidence-based Scoring

Every score should be supported by evidence.

For example:

Completeness: 6/10

The response addresses speaking speed and pauses but does not provide techniques for managing nervousness, which was explicitly mentioned in the user's scenario.

The evaluator should not simply generate arbitrary numbers.

Requirement 3 — Configurable Rubrics

The evaluation criteria should not be permanently hard-coded for one scenario.

The system should allow a rubric such as:

Evaluation Type: Storytelling

Criteria:

- Story structure — 25%
- Relevance — 20%
- Clarity — 20%
- Engagement — 20%
- Completeness — 15%

and another rubric such as:

Evaluation Type: Decision Making

Criteria:

- Problem understanding
- Options considered
- Trade-off analysis
- Reasoning
- Recommendation

The same evaluation framework should be capable of supporting both.

Evaluation Consistency

The evaluator should ideally produce consistent results when evaluating the same response multiple times.

Teams should investigate factors such as:

- model randomness
- temperature
- prompt variations
- rubric wording
- model choice
- scoring calibration

For example:

If the same response is evaluated five times and receives:

8.1
8.0
8.2
8.1
8.0

that is potentially more useful than:

5.8
9.2
7.1
8.9
6.3

Teams should define how they measure and improve evaluation consistency.

Expected Student Deliverables

By the end of the hackathon, teams should deliver the following.

Deliverable 1 — Evaluation Dataset

A small benchmark containing - Use the dataset from elevate as one of the datasets but feel free to come up with different datasets to showcase the different scenarios that the evaluation framework can handle.

- scenarios/tasks
 - AI-generated responses
 - evaluation criteria
 - human-labelled scores
 - examples of strong and weak responses
 - selected edge cases
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Deliverable 2 — Working Evaluation Engine

A functional system that accepts:

Scenario + Response

and produces:

Scores + Evidence + Explanation

Deliverable 3 — Evals Dashboard

Teams can build a simple interface through which a user can view all the training modules generated so far by elevate and their scores:

1. All the scenarios generated
2. Responses

3. Evaluation Rubric
4. View the scores
5. View the evidence and explanation
6. Automatically highlight where the scores are low

The interface does not need to be production-grade.

The focus is on the evaluation engine.

Deliverable 4 — Failure Analysis

Teams must identify situations where their evaluator does **not** perform reliably.

Examples might include:

- Fluent but factually incorrect responses
- Responses that are technically correct but irrelevant
- Very verbose responses receiving artificially high scores
- Creative responses being unfairly penalised
- Different evaluation runs producing significantly different scores
- Evaluator misunderstanding the learner's level
- Evaluator accepting unsupported claims

Understanding these failure modes is an important part of the challenge.
